

A Hydraulic Architecture Based on Multi-common Pressure Rail Principle Using Multi-chamber Cylinders for Excavators



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1 Introduction

The recent trend towards developing sustainable off-road vehicles is pushing towards solutions for the hydraulic actuation system that can increase the overall energy efficiency and reduce fuel consumption. Considering that most of the energy consumption in the off-road vehicle sectors concentrates on agricultural tractors, loaders, and excavators [1], there is high interest towards the development of high-efficient solutions suitable for multi-actuator systems like in excavators. In excavators, the state-of-the-art technology for the actuation system is dominated by centralized systems based on either load sensing solutions or advanced open-center solutions [2]. Research towards more efficient actuation systems for such vehicles has focused on solutions based on higher efficient components, without altering the basic actuation architecture. A significant example is the application of the digital pump concept by Danfoss [3]. However, most of the R&D effort presented in literature pertains to the research of alternative layout architectures. An omni-comprehensive review of promising solutions, some of these also translated to real-market applications, can be found in the review papers [4, 5]. The most energy-efficient hydraulic control architectures reduce throttling losses and allow for energy recuperation during instances of overrunning loads. Among these, there are distributed hydraulic systems (such as displacement control architectures [6]) which increase the number of pumps, as well as centralized architectures based on the concept of common pressure rails, which can retain the cost-effectiveness of current state-of-the-art technology.

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In an MPR system, flow to multiple actuators is provided by several common pressure lines kept at different pressure levels. The flow to each chamber of an actuator is controlled by proportional throttling from the pressure level nearest to the required function pressure, such that the throttling losses can be reduced when multiple actuators need different pressures. Lumkes and Andruch [7] first proposed the idea and used two main supply lines powered by variable displacement pumps and separate meter-in and meter-out valves. By adding accumulators to the pressure rails and integrating optimized engine operations, Vukovic, Sgro, and Murrenhoff [8] improved the architecture and validated the system on a prototype 18-ton excavator [9], which claimed 23.2% reduction in fuel consumption in an air grading cycle comparing to the baseline load sensing (LS) system, showing the outstanding capability of such systems in energy recuperation under overrunning loads frequently seen by construction machines.

The use of multi-chamber cylinder can further increase the flexibility in configuring the connection between pressure rails and cylinder chambers, which uniquely define discrete actuator operation modes. As the actuator has more options to choose as the most efficient mode within certain operating conditions, the throttling losses can be better minimized. Among many studies conducted on the use of multi-chamber cylinders, the most relevant one is the system developed by Heybroek and Sahlman [10], which used four-chamber cylinders with two pressure rails (high and low pressures), which was implemented on a 30-ton excavator and achieved up to 50% fuel efficiency improvement. The integration of MPR systems and multi-chamber cylinders showed noticeable potential regarding improved efficiency by introducing flexibility in the system configurations during operation even prior to the above-cited work. Such flexibility also leads to additional design considerations. Huova, Laamanen, and Linjama [11] showed that proper area selection is needed when multi-chamber cylinders were used to increase system efficiency. In the study carried out by Dell'Amico et al. [12] on controlling a four-chamber cylinder with three-pressure rails as an excavator arm, the trade-off between cylinder accuracy, smoothness, and switching frequency was found by testing different control strategies. The number of pressure rails and cylinder chambers also plays a key role in determining the energy-saving potential. The study by Guo et al. [13], in an agricultural application, shows the efficiency gains per added pressure rail significantly decrease after three rails. Bertolin and Vacca [14] also compared the efficiency differences potentials among different possible configurations of cylinder chambers and pressure rails, considering the case of a mid-size excavator performing a digging cycle. Their findings, which also inspired the present work, can be summarized in Table 1. Essentially, a solution of three-pressure rails and three-chamber cylinders seems to be, at least efficiency-wise, more promising than the case of two pressure rails and four chamber cylinders as considered in prior literature [10]. The three-pressure rail system with three-chamber cylinders consumes 21.5% less energy by including one additional chamber and three extra valves per cylinder, alongside implementing a controller evaluating 16 more operation modes.

Table 1 Analysis on energy consumption differences conducted by Bertolin and Vacca [14]

# Pressure rails	# Cylinder chambers	# Valves per cylinder	# Operation modes	Percent difference [%]
2	2	4	4	+ 57.6
2	3	6	8	+ 9.9
3	2	6	9	Baseline (STEAM [9])
2	4	8	16	−8
3	3	9	27	−21.5
3	4	12	81	−29.3

With one more pressure rail but one less cylinder chamber, such a configuration does not require significantly more hardware compared to the successful implementation by Heybroek and Sahlman [10]. Furthermore, it exhibits promising efficiency improvement at a reasonable cost of increasing controller complexity.

Driven by the above considerations, this paper aims extensively to analyze the design of a three-pressure rail and three-chamber cylinder system, to overcome the possible limitation of the previous study [14]. As a matter of fact, [14] considered the power consumed only by the linear actuators. It also neglected the sizing of the power supply system, so that the analysis could be agnostic with respect to the number of pressure rails. In the presented study, both the engine and the hydraulic pump operation features are taken into consideration, to construct a full simulation model of the overall excavator system. To achieve the best efficiency, a compound controller which estimates the required force or torque and then determines the most efficient operation mode is developed for both the linear and rotary actuators. A pump controller is purposely designed to reduce the pump losses. The overall controller also ensures the engine operation is such that the Brake-Specific Fuel Consumption (BSFC) is minimized. Such model provides simulation results with realistic estimates of the fuel consumption reduction of the proposed solution with respect to a reference baseline where measurement data were available. The simulation results also allow assessing the feasibility of the proposed controllers.

2 Baseline Machine and Duty Cycle

A 22-ton excavator equipped with a negative-flow open center system [2] is taken as reference in the study. The reference cycles considered in this study involve only the four actuators as shown in Fig. 1, although other functions (such as the tracks) are present in the vehicle.

The system essentially separates the four actuators into two circuits to limit load interference, although it includes merging features for the boom and arm systems, where both circuits can supply the actuators. The two supply pumps are usually with

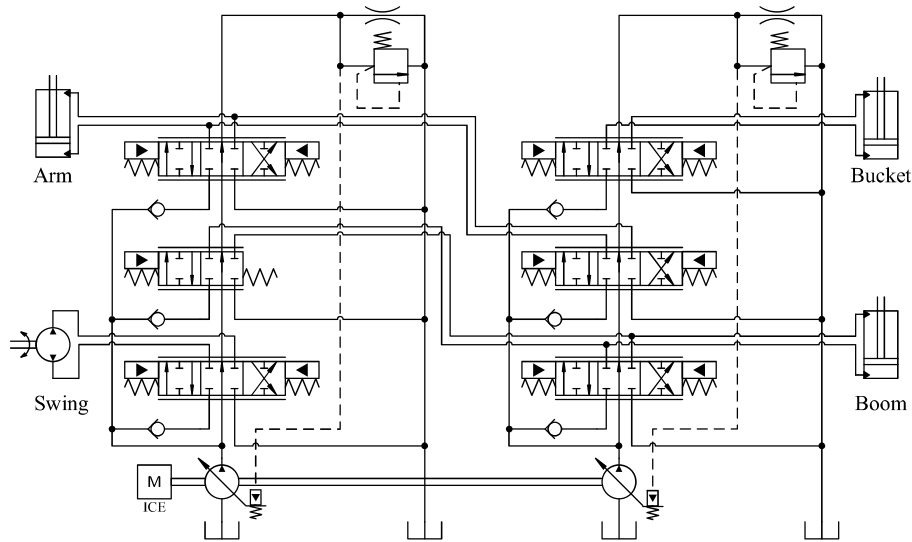


Fig. 1 Simplified schematic of the baseline open center system

the same displacement, to allow meeting the equal flow requirements of the two tracks (not shown in the figure). To evaluate the performance of the simulated system in different conditions, two different duty cycles are studied. The first cycle, named as the “digging cycle”, is a 90° truck loading cycle where the machine rotates 90° from where it starts digging while lifting the boom, and dumps the bucket content into a load receiver. This duty cycle is similar to the one used in [10, 14]. In the second cycle, named as the “grading cycle”, the excavator is used to level and grade the land with empty buckets. The measurements from duty cycles performed by professional operators include cylinder positions, swing velocities, actuator pressures, actuator pilot pressures, engine torque and speed, and pump delivery pressure. These measurements are used as inputs and references for evaluating energy consumption and tracking performance.

3 Hardware Configuration and Simulation Model

The conceptual schematic of the proposed system is presented in Fig. 2. The two variable-displacement pumps in the baseline machine are used to charge one high-pressure (HP) line and one medium-pressure (MP) line, respectively. An additional small fixed-displacement pump is used to charge the low-pressure (LP) line through the integration of a pressure relief valve.

The two variable displacement pumps have equal size as the baseline system, to meet the same high flow demands on both tracks (not shown also in Fig. 2). Each cylinder control manifold consists of nine two-way bi-directional proportional valves which are also capable of load holding. Such a control approach is also applied to the swing motor so that the connections between the motor and the pressure rails can

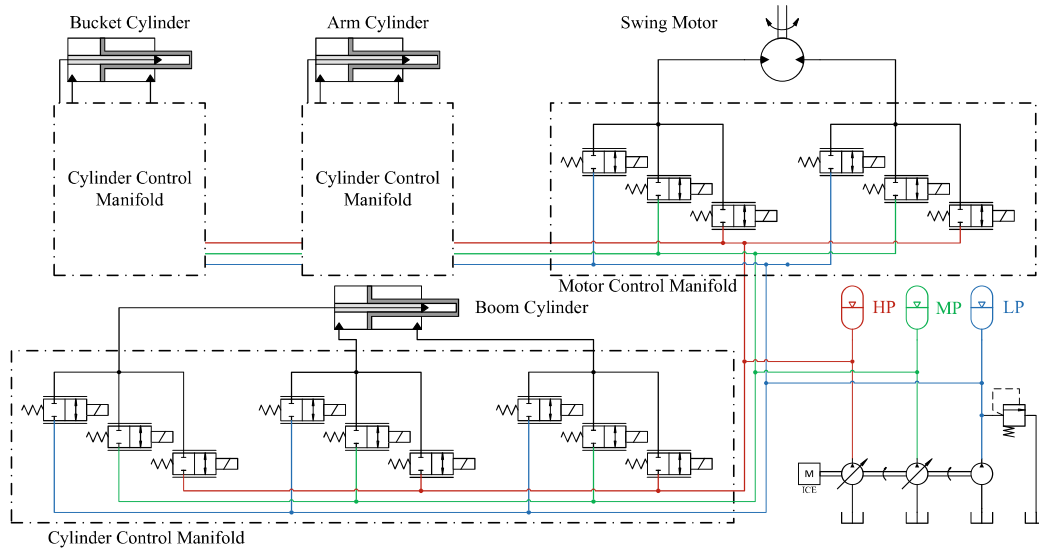


Fig. 2 Conceptual schematic of the proposed system

also be configured independently. The three-chamber cylinders and rail pressures are optimized in a similar manner as proposed in [14].

With the main purpose of evaluating the total power consumption of the proposed system and verifying the feasibility of the designed controllers, a simulation model including the engine operation and all the primary hydraulic components is developed. As shown in Fig. 3, the full simulation model consists of four main sub-system models, including hydraulic system model, engine model, machine dynamics model, and controller. This section describes the first three models which define the dynamics of the excavator, and the controller would be introduced in the next section. All simulation models are developed in MATLAB Simulink.

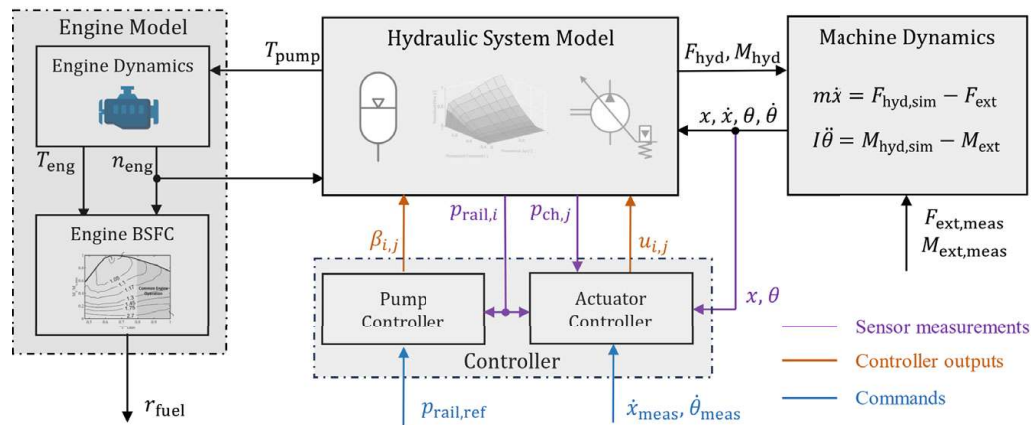


Fig. 3 Conceptual schematic of the simulation model

3.1 Hydraulic System Model

The hydraulic system model is responsible for accounting for all the losses associated with the hydraulic architecture while providing the force or torque generated by the actuators. The primary components in the hydraulic system are pumps, accumulators, cylinders, swing motor, and proportional valves.

3.2 Hydraulic Pumps

To account for the influence of energy losses on the flow rate and the torque at the shaft, the pumps are modeled with empirically derived volumetric efficiencies and hydromechanical efficiencies that vary with pressure drop, displacement, and shaft speed.

The dynamics of the pump normalized displacement is also taken into consideration by a first-order transfer function as

$$\dot{\beta} = \frac{1}{\tau_p} \beta_{\text{cmd}} - \frac{1}{\tau_p} \beta \quad (1)$$

3.3 Accumulators

With the assumption of ideal gas and adiabatic processes and the oil pressure forced to be equal to the gas pressure, the resulting hydraulic capacitance of the accumulator is

$$C_{\text{acc}} = \frac{V_g}{\gamma} \left(\frac{p_g}{p_{\text{oil}}^{1+\gamma}} \right)^{\frac{1}{\gamma}} + \frac{V_{\text{oil}}}{B_T} \quad (2)$$

The calculated hydraulic capacitance is then applied to the pressure build-up equation to model the accumulator pressure dynamics.

3.4 Cylinders

Ignoring the internal leakage between chambers, the pressure build-up equation, as derived in [2], is applied to evaluate the pressure in each chamber of the cylinders.

With the pressures in each chamber defined, the hydraulic force generated by the cylinder can be calculated using cross-sectional areas.

3.5 *Swing Motor*

The swing motor is modeled alongside the pressure dynamics of the hydraulic lines around it by pressure build-up equations. Since the motor can operate in either positive or negative speeds with either positive or negative pressure differential, the logic to use the volumetric efficiency and hydromechanical efficiency of hydrostatic units needs to be modified according to operation quadrants.

3.6 *Proportional Valves*

To predict the throttling losses as accurately as possible, the pressure-flow characteristics of the proportional valves are implemented based on steady-state performance curves provided by the manufacturer. Thus,

$$Q_{\text{prop}} = \text{sgn}(\Delta p) \cdot f(y_{\text{prop}}, |\Delta p|) \quad (3)$$

The dynamics of the proportional valves relative to an electronic control signal are modeled by a first-order system as

$$\dot{y}_{\text{prop}} = \frac{1}{\tau_{\text{prop}}} u_{\text{prop}} - \frac{1}{\tau_{\text{prop}}} y_{\text{prop}} \quad (4)$$

The response time of such valves is assumed to be 100 ms, which is reasonable for such proportional valves available in the market.

3.7 *Engine Model*

The engine model is used to evaluate the fuel consumption based on engine torque and engine speed in operations, as well as to account for the speed dynamics and torque limitations. The engine speed dynamics is obtained with torque balance in the engine shaft, described as

$$I_{\text{eng}} \dot{\omega}_{\text{eng}} = M_{\text{eng}} - M_{\text{L}} - b_{\text{eng}} \omega_{\text{eng}} \quad (5)$$

To model the dynamics of the generated engine torque, an approach similar to the one proposed by Dekraker, Stuhldreher, and Kim [15] is used. A PI controller tracking a reference is used to model the engine governor and to output the engine torque command, which is then saturated by the maximum the engine can provide at the given engine speed. Subsequently, two first-order transfer functions are applied to the saturated torque to model the delay in engine torque generation.

To validate the engine model, the measured pump pressures, flow rates, and engine shaft speeds are input into the pump model described in the previous section, which calculates the effective load torque at the engine. The engine model parameters are then optimized in such a way that the simulated speed profile fits the measured engine speed with the calculated load torque.

The fuel consumptions are evaluated with the optimized model and BSFC data in steady state for both the baseline system and the proposed system. The BSFC is given by

$$\text{BSFC} = \frac{r}{P} = \frac{r}{M_{\text{eng}} \cdot n_{\text{eng}}} \quad (6)$$

The BSFC map with respect to engine torque and speed is provided by the engine manufacturer which is covered by confidentiality.

3.8 Machine Dynamics Model

The machine dynamics model is developed to simulate the movement of the actuators and to check the position/velocity tracking performances. Instead of using the machine kinematic information to predict the equivalent mass/inertia with actuator positions and estimating the friction forces based on empirical evaluations, the total external force/torque acting on each actuator is approximated directly through measurements. Take the case of linear actuators as an example, the following equation holds during measurements:

$$m\ddot{x}_{\text{cyl, meas}} = F_{\text{hyd, meas}} - F_{\text{L}} - F_{\text{f}}(\dot{x}_{\text{cyl}}). \quad (7)$$

The sum of the load force and the friction force can be calculated as

$$F_{\text{ext}} = F_{\text{load}} + F_{\text{f}}(\dot{x}_{\text{cyl}}) = -F_{\text{hyd, meas}} - m\ddot{x}_{\text{cyl, meas}}. \quad (8)$$

Then the calculated total external force can be input to the equation of motion of the cylinder piston in simulation:

$$m\ddot{x} = F_{\text{hyd, sim}} - F_{\text{ext}}. \quad (9)$$

The key idea of such approximation is to ask the generated cylinder force to follow the hydraulic force in measured duty cycles, while also keeping well track of the actuator positions and velocities. In this way, a relatively fair energy comparison is attained by forcing the power output by actuators as close as possible. On one hand, such an approach can be used for energy evaluations of a given duty cycle with measurements available. On the other hand, it cannot be used to predict the performance of any arbitrary duty cycles due to lack of a kinematic model. It also needs to be mentioned that this approach assumes the same friction force for different cylinder designs.

4 Controllers

Since charging of pressure rails and controlling actuators through throttling from pressure rails are almost fully decoupled, it is possible to design the pump displacement controllers and the actuator controllers so that both the pumps and the actuators operate with the best efficiency. On the one hand, the design objective of the pump controller is to minimize fuel consumption while ensuring power availability to the actuators. On the other hand, the design objective of the actuator controller is to minimize the total power taken from the pressure rails while maintaining the tracking performance.

4.1 *Pump Controller and Modified Engine Operation*

To minimize the total fuel consumed for charging the pressure rails, both reducing pump losses and increasing engine fuel efficiencies are taken into account. As pump efficiencies are maximized with full displacements, it is preferred to set the pumps either in charging mode with maximal available displacements or standby mode with minimum displacements to account for the leakage due to pressure drop across the pump. To maintain a fairly constant power demand and to avoid frequent part loading which leads to inefficient use of the engine, the engine is set to operate in the highest efficiency range with a lower reference shaft speed. Thus, the pumps tend to charge longer with the lower shaft speed and unload the engine less frequently. Moreover, a torque limiter is needed to avoid engine stalling because the torque of operating both pumps at full displacements may exceed the maximum torque that is available from the engine. Since the HP rail ensures power availability to the system, the displacement of MP pump would be saturated according to available engine torque.

4.2 Actuator Controller

The actuator controller consists of four main components: a velocity controller, a mode selector, a pressure controller, and a flow allocator. An overview of the controller concepts is presented in Fig. 4.

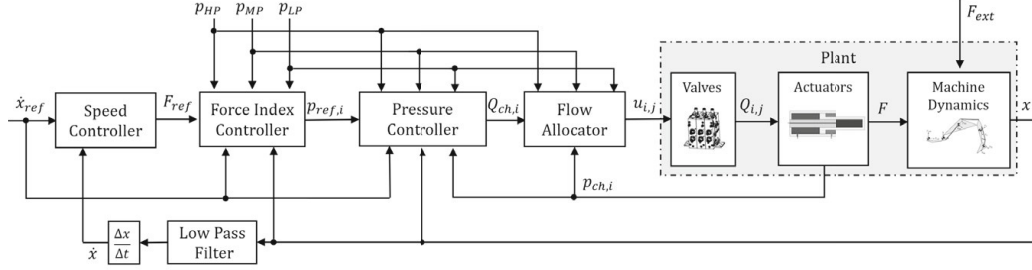


Fig. 4 Overview of the actuator controller structure

4.3 Speed Controller

Similar to the concept of secondary control, closed-loop velocity control is utilized for the actuators. Different from the controller developed by Heybroek and Sahlman [10] which only uses the machine model to generate the reference force, an additional adaptive speed controller is implemented to assist speed tracking under changing loads depending on the actuator positions. A similar secondary control approach has been successfully implemented by Busquets and Ivantysynova [6] on rotary actuators. As the focus of the paper is to evaluate the overall performance of the architecture on the full machine level, the details of the derivation of the adaptive controller are out of scope and are not included.

4.4 Force Index Controller

Once the reference force is determined, the best operation mode can be selected based on certain criteria evaluating each mode in terms of energy efficiency and force tracking. An approach similar to the Force Index Controller (FIC) proposed by Heybroek and Sjöberg [16] is used but with a different cost function. In addition to the first term evaluating the throttling losses of each mode

$$J_1 = k_1 \cdot |v_{cyl} \cdot (F_{ref} - F_{mode})|, \quad (10)$$

and the second term evaluating the compressibility losses due to switching

$$J_2 = k_2 \cdot \sum_{i=1}^3 p_i \cdot - (p_{i,\text{new}} - p_i) \cdot \frac{V_i}{B} \quad (11)$$

ΔV_i

a term penalizing the number of chambers whose connection with the rails are to be changed

$$J_3 = \begin{cases} k_3 \cdot \#sw^4, & \#sw \leq 2 \\ \infty, & \#sw = 3 \end{cases} \quad (12)$$

and a term evaluating the switching loss due to cross leakage between pressure rails

$$J_4 = k_4 \cdot \sum_{i=1}^3 (p_{i,\text{rail,new}} - p_{i,\text{rail}})^{3/2}, \quad (13)$$

are added to better optimize the mode selection and tradeoff between minimizing throttling losses and reducing switching frequency. The coefficients are manually optimized based on the performance in the two given duty cycles.

4.5 Pressure Controller

With the given desired pressure in each chamber, an adaptive model-predictive controller is used to find the optimal pressure trajectories during transient to minimize the force tracking error, thus smoothing the dynamic performances during switching. The desired flow rate Q_{ref} for each chamber is determined by minimizing

$$J_{\text{MPC}} = \sum_{t=1}^N W^F (F_{\text{ref}} - A^T \cdot \varphi \cdot p_{\text{ch},t})^2 + (p_{\text{ch,ref}} - p_{\text{ch},t})^T W^p (p_{\text{ch,ref}} - p_{\text{ch},t}) \\ + (Q_{\text{ref},k} - Q_{\text{ref},k-1})^T W^{\Delta Q} (Q_{\text{ref},k} - Q_{\text{ref},k-1}), \quad (14)$$

where the first term penalizes force tracking error, the second term penalizes pressure tracking error, and the last term penalizes large changes in desired flow rates as control inputs. Upper bounds and lower bounds on the flow rates are implemented to ensure that the commanded flow rates are available through the proportional valves. The MATLAB MPC toolbox is used for implementing the controller.

4.6 Flow Allocator

Based on the flow rates commanded by the pressure controller and chamber pressures, the flow allocator calculates valve opening commands to get enough flow from pressure rails. While it is always possible to get enough flow from HP line or to send enough flow to LP line, the logic is implemented such that the flow from or going to nearest pressure lines is always preferred. Such logic is primarily used in transient states when changing operating modes, while at steady state the pressure source for each chamber would be determined by the pressure distribution optimized by the Force Index Controller. An inverse map provided by the valve manufacturer is used to derive the required normalized opening commands for a given flow rate request and pressure differential.

5 Simulation Results

The model and controller described above are used to perform the simulation of the two reference duty cycles. It is essential to ensure similar delivered power from actuators when energy consumption is in interest. Instead of inputting the measured operator commands and compare the resulting performances, which might lead to different actuator position profiles and delivered power, the controllers are utilized to back calculate the valve commands for tracking the measured positions under external forces calculated from measurements.

5.1 Cycle Tracking

The simulated results showing the position tracking performances in the grading cycle are presented in Fig. 5. Despite the tracking errors due to estimating equivalent loads for each actuator based on tracking performances, the simulated trajectories show a good match with the measurements.

Force tracking and transient performances during mode switches are shown in Fig. 6, where the simulated force follows well the reference commanded by the velocity controller. The short spikes during changing forces modes are considered to be acceptable given the large inertia of cylinder pistons. The result shows the controller is able to change operations modes for reduced throttling losses when external forces change.

Verifying the cycle tracking performance not only confirms the feasibility of the developed controller but also backs fair energy consumption comparisons.

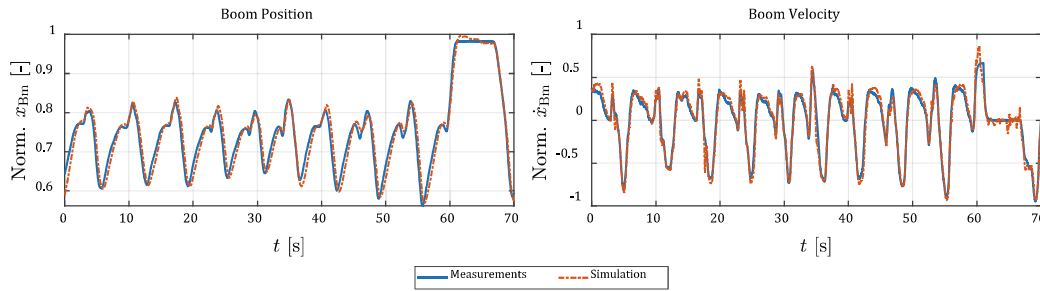


Fig. 5 Position and velocity tracking with normalized properties in the grading cycle

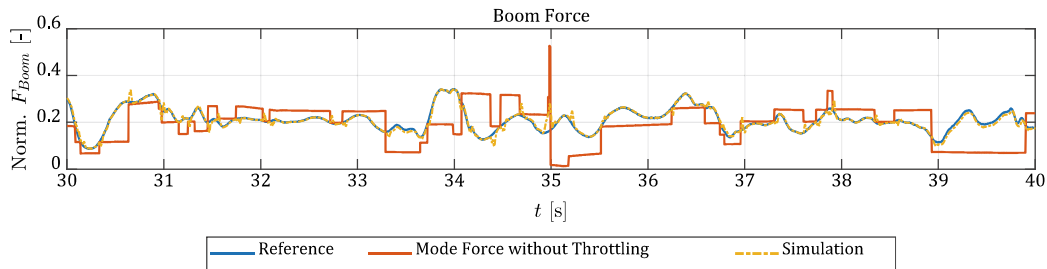


Fig. 6 Boom force tracking performance with normalized properties in the grading cycle

5.2 Power Supply Performance

The pressure dynamics in accumulators and corresponding pump operations are presented in Fig. 7. While the pumps are controlled simply to maintain pressures within a certain range, there are moments when actuators are taking more flow than a single pump can provide. Nevertheless, low pressure due to the excessive flow command only occurs when both the arm and boom cylinders are pushed to maximum position rapidly, after which maintaining high pressures in the high-pressure rail for potential high power output is no longer necessary.

The pump displacements show that the HP pump is frequently charging the pressure rail and the MP pump is always trying to charge the accumulator under the torque limits. This means the system is operating close to its maximum capacity. Another important thing to be noticed is the similar accumulator pressures in the beginning and the end of the duty cycles, which reflects that the energy consumption evaluation does not take any advantage of the power stored in the accumulator.

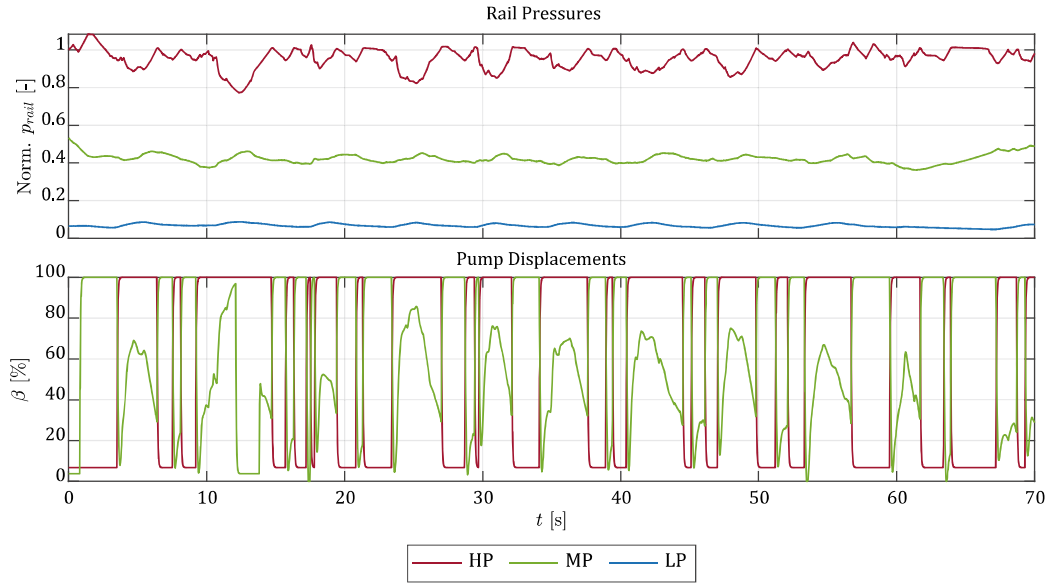


Fig. 7 Power supply performances with normalized properties in the grading cycle

5.3 Engine Operation

The high charging frequency leads to less partial load operation of the engine, as shown in Fig. 8. The contours represent the fuel efficiencies evaluated by BSFC within maximum torque. The time spent at each operation point is indicated by the histogram. With modified speed optimized for the given two duty cycles, the engine can operate in the desired region with the best fuel efficiency throughout the majority

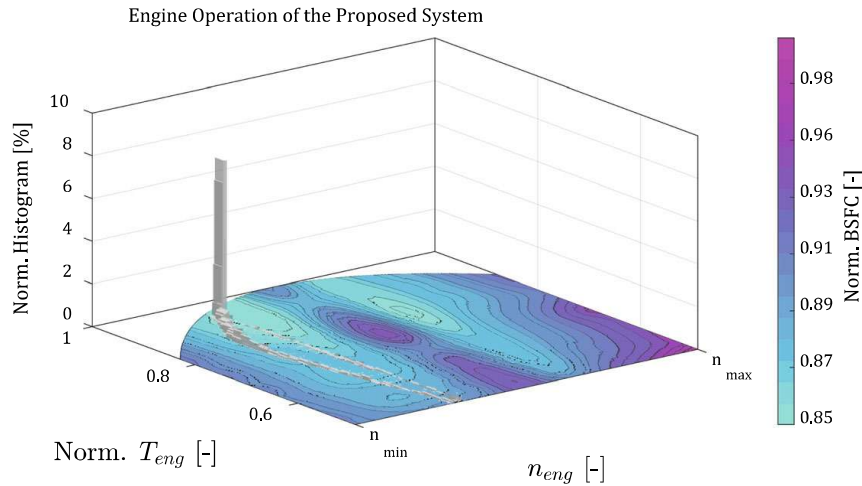


Fig. 8 Engine operations of the proposed system with normalized properties in the grading cycle

Table 2 Percentage difference of the proposed system in average power, BSFC, and total fuel consumption when compared to the baseline system

Duty cycle	Average Power	Average BSFC	Fuel Consumption
Digging	−18.8%	−5.1%	−22.3%
Grading	−34.3%	−2.2%	−36.0%

of the cycle duration. This leads to higher fuel efficiencies than the baseline engine operations, where a higher engine speed is maintained with frequent partial loading.

5.4 Fuel Consumption Evaluations

With the previous sections showing the evidence of fair energy consumption comparisons and intuitively more efficient engine operations, Table 2 summarizes the comparisons between the proposed system and the baseline system in average power, average BSFC, and total fuel consumption in the two duty cycles.

The results indicate that the proposed structure can reduce the fuel consumption by 22.3% in the digging cycle and 36.0% in the grading cycle when compared to the baseline machine. Due to the improved engine efficiency, the final fuel savings are even slightly higher.

It is interesting to observe the differences in efficiency gains between two duty cycles. The differences reveal the different loss distributions in the two duty cycles. Depending on operator behaviors, the energy that can be restored and the throttle losses that can be reduced are distinct in various cycles.

6 Conclusions

A hydraulic architecture based on the concept of MPR systems and multi-chamber cylinders, with the specific configuration of three-pressure rails and three-chamber cylinders, was proposed and simulated in the case of a mid-size excavator. The system uses two variable displacement pumps and one small, fixed displacement pump to charge three accumulators, which are viewed as pressure rails and connected to all primary actuators. Parallel installed proportional valves enable flexible connections between pressure rails and actuators. A compound controller aiming at tracking given actuator speed command with maximized efficiency was developed. The pump controller and engine operations were also modified to reduce pump losses and partial loading conditions.

Two different duty cycles were simulated to evaluate the fuel consumption and to assess the controllability. With competent cycle tracking performances as well as capability of changing operation modes for optimized efficiency, results showed a fuel

consumption reduction up to 36%, depending on the cycles. The reduction is mainly due to reduced power requirements from the hydraulic system, while improved engine efficiency also contributes.

The next step would be to further reason the different efficiency gains in different cycles for a better understanding of whether better performances are possible given the work cycle and hardware configurations.

Abbreviations

Nomenclature

β	Normalized pump displacement
τ	Time constant
C	Equivalent hydraulic capacitance
p	Pressure
V	Volume
B_T	Isothermal bulk modulus
γ	Polytropic index
Q	Flow rate
Δp	Pressure differential
y	Position of valve poppet
u	Opening area of valves
I	Equivalent rotational inertia
n/ω	Shaft speed
M	Shaft Torque
b	Empirical friction coefficient
r	Fuel consumption rate
P	Produced power by the engine
m	Mass
x	Position
F	Force
J	Penalty term
k	Weight of penalty term
v	Velocity
$\#sw$	Number of switches
k	Weight of penalty term
W	Weight matrix
φ	Force direction coefficient
A	Cross-sectional area

Subscripts

P	Pump
cmd	Command
acc	Accumulator
mech	Hydromechanical
g	Pre-charged gas in accumulators
oil	Oil in accumulators
prop	Proportional valve
eng	Engine
L	Load
cyl	Cylinder
meas	Measurement
hyd	Hydraulic
f	Friction
ext	Sum of all external forces
sim	Simulation
ref	Reference
mode	Operation mode without throttling
rail	Pressure rail
<i>i</i>	Chamber index
<i>j</i>	Pressure rail index
HP	High-pressure rail
MP	Medium-pressure rail
LP	Low-pressure rail

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